

Calculus I Lesson 41

1) Evaluate $\int \tan x \cdot dx$.

Hint: $\tan x = \frac{\sin x}{\cos x} = \frac{1}{\cos x} \cdot \sin x$

a) Let $u = \cos x$. $\frac{du}{dx} = \underline{\hspace{1cm}}?$ and $-1 \cdot du = \underline{\hspace{1cm}}?$

b) Write the indefinite integral $\int \tan x \cdot dx = \int \frac{1}{\cos x} \cdot \sin x \cdot dx$ in terms of u and then evaluate.

$$\int \frac{1}{\cos x} \cdot \sin x \cdot dx = \int (\underline{\hspace{1cm}}?) \cdot du = \underline{\hspace{1cm}}?$$

Hint: $\int \frac{1}{u} \cdot du = \ln|u| + C$

c) $\int \tan x \cdot dx = \underline{\hspace{1cm}}?$ (Note: Answer must be in terms of x .)

2) Evaluate $\int \cot x \cdot dx$.

Hint: $\cot x = \frac{\cos x}{\sin x} = \frac{1}{\sin x} \cdot \cos x$

a) Let $u = \sin x$. $\frac{du}{dx} = \underline{\hspace{1cm}}?$ and $du = \underline{\hspace{1cm}}?$

b) Write the indefinite integral $\int \cot x \cdot dx = \int \frac{1}{\sin x} \cdot \cos x \cdot dx$ in terms of u and then evaluate.

$$\int \frac{1}{\sin x} \cdot \cos x \cdot dx = \int (\underline{\hspace{1cm}}?) \cdot du = \underline{\hspace{1cm}}?$$

Hint: $\int \frac{1}{u} \cdot du = \ln|u| + C$

c) $\int \cot x \cdot dx = \underline{\hspace{1cm}}?$ (Note: Answer must be in terms of x .)

3) Evaluate $\int \sec x \cdot dx$.

Hint: $\sec x = \sec x \cdot \frac{(\sec x + \tan x)}{(\sec x + \tan x)} = \frac{\sec^2 x + \sec x \cdot \tan x}{\sec x + \tan x} = \frac{1}{\sec x + \tan x} \cdot (\sec^2 x + \sec x \cdot \tan x)$

a) Let $u = \sec x + \tan x$. $\frac{du}{dx} = \sec x \cdot \tan x + \sec^2 x$ and $du = \underline{\hspace{2cm}}?$

b) Write the indefinite integral $\int \sec x \cdot dx = \int \frac{1}{\sec x + \tan x} \cdot (\sec^2 x + \sec x \cdot \tan x) \cdot dx$
in terms of u and then evaluate.

$\int \sec x \cdot dx = \int (\underline{\hspace{2cm}}?) \cdot du = \underline{\hspace{2cm}}?$ Hint: $\int \frac{1}{u} \cdot du = \ln|u| + C$

c) $\int \sec x \cdot dx = \underline{\hspace{2cm}}?$ (Note: Answer must be in terms of x .)

3) Evaluate $\int \sec x \cdot dx$.

Hint: $\sec x = \sec x \cdot \frac{(\sec x + \tan x)}{(\sec x + \tan x)} = \frac{\sec^2 x + \sec x \cdot \tan x}{\sec x + \tan x} = \frac{1}{\sec x + \tan x} \cdot (\sec^2 x + \sec x \cdot \tan x)$

a) Let $u = \sec x + \tan x$. $\frac{du}{dx} = \sec x \cdot \tan x + \sec^2 x$ and $du = \underline{\hspace{1cm}}?$

b) Write the indefinite integral $\int \sec x \cdot dx = \int \frac{1}{\sec x + \tan x} \cdot (\sec^2 x + \sec x \cdot \tan x) \cdot dx$
in terms of u and then evaluate.

$\int \sec x \cdot dx = \int (\underline{\hspace{1cm}}?) \cdot du = \underline{\hspace{1cm}}?$ Hint: $\int \frac{1}{u} \cdot du = \ln|u| + C$

c) $\int \sec x \cdot dx = \underline{\hspace{1cm}}?$ (Note: Answer must be in terms of x .)

5) Evaluate $\int \csc(5x - 4) \cdot dx$.

a) Let $u = 5x - 4$. $\frac{du}{dx} = 5$ and $\frac{1}{5} \cdot du = \underline{\hspace{1cm}}?$

b) Write the indefinite integral $\int \csc(5x - 4) \cdot dx$ in terms of u and then evaluate.

$$\int \csc(5x - 4) \cdot dx = \int (\underline{\hspace{1cm}}?) \cdot du = \underline{\hspace{1cm}}? \quad \text{Hint: Use the result from Problem 4}$$

c) $\int \csc(5x - 4) \cdot dx = \underline{\hspace{1cm}}?$ (Note: Answer must be in terms of x .)

6) Find: $\int \cot\left(\frac{4x}{3}\right) dx$

Hint: Let $u = \frac{4x}{3} = \frac{4}{3}x$

Hint: $\int \frac{1}{x} dx = \ln|x| + C$ $\int \frac{1}{x \pm b} dx = \ln|x \pm b| + C$ $\int \frac{1}{ax \pm b} dx = \frac{1}{a} \ln|x \pm b| + C$

a) $\frac{du}{dx} = \underline{\hspace{2cm}}?$

b) $du = \underline{\hspace{2cm}}?$

c) $\int \cot\left(\frac{4x}{3}\right) dx = \underline{\hspace{2cm}}?$

7) Find: $\int \csc(14x) dx$

Hint: Let $u = 14x$.

Hint: $\int \frac{1}{x} dx = \ln|x| + C$ $\int \frac{1}{x \pm b} dx = \ln|x \pm b| + C$ $\int \frac{1}{ax \pm b} dx = \frac{1}{a} \ln|x \pm b| + C$

a) $\frac{du}{dx} = \underline{\hspace{2cm}}?$

b) $du = \underline{\hspace{2cm}}?$

c) $\int \csc(14x) dx = \underline{\hspace{2cm}}?$

8) Find: $\int (\cos 5x - 10)dx = \int (\cos 5x)dx - \int (10)dx.$

Hint: For $\int (\cos 5x)dx$, let $u = 5x$

a) $\int (\cos 5x)dx = \underline{\hspace{2cm}}?$

b) $\int (10)dx = \underline{\hspace{2cm}}?$

c) $\int (\cos 5x - 10)dx = \underline{\hspace{2cm}}?$

